

September 18, 2026

1 Material Balances I

What is Bond listening to right now?

This Notebook contrasts intensive and extensive properties of systems, and it introduces material balances on reacting systems.

1.1 Intensive and Extensive Properties

It is worth a brief discussion of intensive properties and extensive properties. We have to relate the two frequently in reactor design, so it is important that we have good intuition about what they are and how they are different. Below, we'll establish a working conceptual definition of these two things:

1.1.1 Intensive Properties

Intensive properties are scale-independent. They are presented as values per unit of something (e.g., per unit volume, per unit mass, per unit area, etc.). Some examples would be concentration (moles per volume); density (mass per volume); mole fraction (moles per total moles); etc. We also typically report reaction rates as intensive quantities that are normalized to volume, catalyst mass, or to some other characteristic dimension of the system. Intensive properties are the same whether you're describing a large reactor or a small reactor. For example, a 1 Molar solution of sulfuric acid always has a concentration of 1 mole H_2SO_4 per liter of solution volume whether you are considering that sulfuric acid in a 1L system or a 100L system. Intensive properties are extremely useful because they are system/scale independent.

1.1.2 Extensive Properties

Extensive Properties are scale-dependent. They will not be presented as being normalized to a unit value of something. Some examples include total moles in a reactor, molar flowrates of species entering and leaving a reactor, volumes of reactors, and the mass of catalyst inside of a reactor. These quantities change as you consider processes of different sizes or scales. They are critically important because they define the actual physical size of our system, throughput, revenue, cost, and profitability.

A good test to determine whether a property is intensive or extensive is to ask "If I double the size of the system, or if I cut the size of the system in half, does the value of this property change?" If it does not change, it is an intensive property. If it does change, it is an extensive property.

1.1.3 The Relationship Between Intensive and Extensive Properties

For an illustration of intensive and extensive properties, consider a 1L tank filled with a 1 molar solution of species A , i.e., $C_A = 1$ mole/liter. If we wanted to know the total moles of A in that tank, we would calculate it as:

$$N_A = C_A V$$

So for this case, we would have 1 mole of A in our 1L reactor. If we take the same solution and fill a 100L reactor with it, we would find that there are 100 total moles of A in the 100L reactor. The concentration of A , C_A , is the same in both cases (1M). Concentration is an intensive variable, and it is independent of scale (the size of the system). The total number of moles of A , N_A , is an extensive variable. As we fill larger and larger tanks with a 1M solution of A , we will get larger and larger quantities of moles of A . If you get this basic concept, this is an adequate working understanding of intensive and extensive properties and how both might show up in a reactor design problem.

1.2 How do reactors scale anyway?

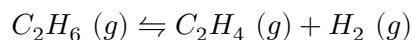
Our courses have primarily conditioned us to think of intensive properties in units per volume, i.e., they are properties specified per unit volume of a mixture. We also usually think about the size of the reactor as a volume, but it is important to consider that the *extensive productivity* of a reactor (e.g., how many moles it produces annually) does not necessarily scale closely with the volume of the reactor. Sometimes, it will scale more strongly with the mass of catalyst in the reactor, the interfacial area between two separate phases, or the number of accessible catalytic active sites in our system. To understand the critical “size” dimension of a reactor, we should first consider the differences between homogeneous reactions and heterogeneous reactions.

In a **Homogeneous Reaction** all species—reactants, intermediates, products, and/or catalysts—are present in a single phase. Usually, this will be either a gas or a solution phase. Note that when we say solution, we mean a single liquid phase wherein all species are miscible and/or fully soluble.

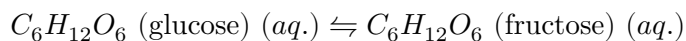
In a **Heterogeneous Reaction** at least one species—reactant, intermediate, product, or catalyst—is present in a second phase. Note that two separate liquid phases can be an example of a heterogeneous reaction.

1.2.1 Homogeneous Reactions

One example of a homogeneous reaction is ethane cracking, which is the dominant process for converting ethane from oil and gas wells into the far more valuable commodity chemical, ethylene:



This reaction is non-catalytic. It occurs in the gas phase. The mechanism of this reaction involves the formation of radical intermediates, which are also present in the gas phase. Another example of a liquid-phase homogeneous reaction is glucose isomerization:



This reaction occurs in aqueous solution (i.e., in water), and it uses a water-soluble enzyme (xylose isomerase) as a catalyst. All of the intermediates that form in the glucose isomerization mechanism are in aqueous solution, so we classify this as a homogeneous reaction. An important thing to recognize about a homogeneous reaction is that it occurs *within the fluid*. As a consequence, it is happening throughout the space (volume) occupied by the fluid. One can then imagine that the *extensive* quantities of species produced in a homogeneous reaction will scale with the *volume* of fluid that is reacting.

A thought experiment As an example, let's say we study ethane pyrolysis to produce ethylene ($C_2H_6(g) \rightleftharpoons C_2H_4(g) + H_2(g)$) in a small scale batch reactor that has a volume of 1L. Under these conditions, the rate of reaction is constant at 1 mole of extent per liter per minute (this will not normally be true, but we will assume it here to keep the thought experiment simple). We fill that reactor with 1 mole of ethane so that it has a concentration of 1M. The ethane will expand to fill the entire reactor volume, and the reaction occurs throughout the volume occupied. If we assume the reaction is irreversible, and we let this system react for one minute, we would make 1 mole of ethylene (C_2H_4) and 1 mole of H_2

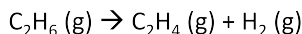
Now, let's consider this same reaction in a 1000L reactor, and we run it under the same temperature, pressure, and intensive condition (i.e., an ethane concentration of 1M). To do this, we would initially charge this 1000L reactor with an extensive quantity of 1000 moles of ethane (C_2H_6) so that the ethane concentration is 1M. The reaction again occurs throughout the entire volume of the reactor, and we again have the same, constant intensive reaction rate of 1 mole of extent per liter per minute. If we let this system react for 1 minute in this 1000L reactor, we would form 1000 moles of each product. Contrast this to the case of the 1L reactor, which produces only 1 mole of product under identical conditions. In this case, as we increase the volume of the reactor, the reaction media fills that volume. The extensive productivity of the reactor (total moles produced) increases accordingly, whereas intensive properties (rate per volume, concentration of ethane) are identical in the two systems because they are scale independent.

For a Homogeneous Reaction:

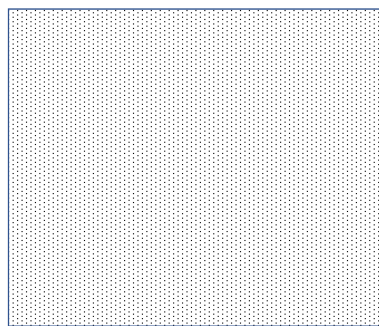
Reaction occurs throughout volume
Volume is a good description of "size"

Example:

Ethane Pyrolysis



1L Reactor
1 Mole of ethane (dots)
 $C_E = 1$ mole/L
Ethane fills reactor space
 $r = 1$ mol/L/min
 $t = 1$ minute
Produces 1 mole of each product



1000L Reactor
1000 Moles of ethane (dots)
 $C_E = 1$ mole/L
Ethane fills reactor space
 $r = 1$ mol/L/min
 $t = 1$ minute
Produces 1000 moles of each product

Because *homogeneous* reactions tend to occur throughout the space occupied by the fluid phase, reason, when we consider *homogeneous* reactions, we tend to think of the reactor size in terms of the volume occupied by the fluid phase.

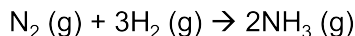
1.2.2 Heterogeneous Reactions

Heterogeneous reactions are much more varied. We can imagine lots of cases where a second phase of matter is present. A classic one is a gas-phase or liquid-phase reaction catalyzed by a solid catalyst. We can also imagine a reaction between two immiscible liquids (See Equilibrium Examples from [Notebook 12](#)), or a chemical vapor deposition process, wherein gas-phase species react to form solid products. We'll go through a few examples below.

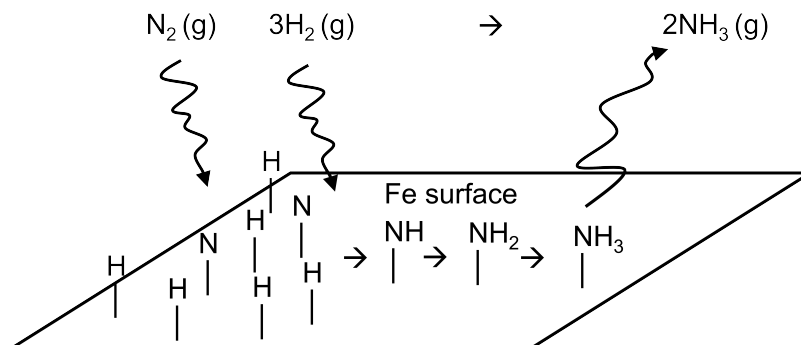
Catalytic Reactions at Surfaces: Ammonia Synthesis Superficially, this looks like a gas-phase reaction, but it is actually catalyzed by (and occurring on) a solid surface. So, here, we have gas phase reactants (N_2 and H_2) and a gas phase product (NH_3), but we have a solid catalyst (usually an Iron-based material), and we have intermediates that are chemically adsorbed onto the atoms on that metal surface. So, in this case, the catalyst and the intermediates are all present as a solid phase, and we have a heterogeneous reaction.

Ammonia Synthesis:

An example of Heterogeneous Reaction



This reaction occurs over Fe-based catalysts



Here, the reaction is *only* occurring on the catalyst. It does not occur throughout the space occupied by the fluid, i.e., the reactor volume. In this case, the critical component of the reactor that dictates its extensive productivity is the quantity of catalyst inside of the reactor, not the volume of the reactor itself. Often, it is convenient to quantify amount of catalyst in terms of its mass; therefore, mass of catalysis is a common scaling dimension for reactions that occur on solid surfaces. For an illustration, consider the thought experiment below.

Scaling a Heterogeneously catalyzed reaction This is a relatively loose conceptual illustration because it neglects the dependence of reaction rate on species concentration, but it gets to the point about how the definition of “size” of a system really depends on the type of reaction we’re

considering. In reality, changing the volume of a catalytic reactor will usually have an impact on the conversion achieved in that reactor, it just won't be as significant as changing the amount of catalyst inside of the reactor.

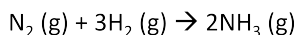
For a Heterogeneously Catalyzed Reaction:

Reaction occurs only on catalyst

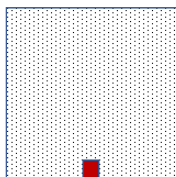
Mass of catalyst is a better description of "size"

Example:

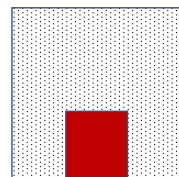
Ammonia Synthesis



1L Reactor
1 Mole of N_2 (gas)
3 Moles of H_2 (gas)
1 gram of catalyst (red)
Reaction occurs on catalyst
 $r' = 1 \text{ mol/min/g}$
 $t = 1 \text{ minute}$
Produces 2 moles NH_3



10L Reactor
10 Moles of N_2
30 Moles of H_2
1 gram of catalyst
Reaction occurs on catalyst
 $r' = 1 \text{ mol/min/g}$
 $t = 1 \text{ minute}$
Produces 2 moles of NH_3



10L Reactor
10 Mole of N_2 (gas)
30 Moles of H_2 (gas)
10 grams of catalyst (red)
Reaction occurs on catalyst
 $r' = 1 \text{ mol/min/g}$
 $t = 1 \text{ minute}$
Produces 20 moles NH_3

Imagine we mix 1 mole of N_2 and 3 moles of H_2 in a 1L reactor ($C_{\text{N}_2} = 1M$ and $C_{\text{H}_2} = 3M$), and we add 1 gram of catalyst. This catalyst achieves a constant *intensive* rate of reaction of 1 mole of extent per gram of catalyst per minute. For simplicity, we are assuming that the rate of reaction is independent of species concentration so that the rate of reaction is constant throughout the reaction progress. This is not always going to be true. We allow this system to react for one minute, at which point all of the N_2 and H_2 are consumed. At this point in time, we will have produced 2 moles of NH_3 in the gas phase ($C_{\text{NH}_3} = 2M$).

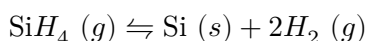
Now we want to scale this process. In the second illustration of the above figure, we increase the volume of the reactor and keep the initial gas phase concentrations of N_2 and H_2 constant. We also keep 1 gram of catalyst constant; that catalyst still achieves a rate of reaction of 1 mole of extent per gram of catalyst per minute; and we allow the system to again react for 1 minute. In this case, since the reaction only occurs on the catalyst, and we have kept the quantity of catalyst the same (1 gram), we find that the total volume of the system is irrelevant to determining the total quantity of N_2 and H_2 that are converted into ammonia. After 1 minute elapses, we still consume 1 mole of N_2 and 3 moles of H_2 , and we still produce 2 moles of NH_3 . We won't increase the extensive quantity of ammonia produced (the scale of the process) until we increase the amount of catalyst, which is shown in the 3rd illustration. Only when we increase the quantity of catalyst to 10 grams would we convert 10 moles of N_2 into 20 moles of NH_3 in one minute.

Because of this scaling, for a reaction catalyzed by a solid, we often see quantities reported per mass of catalyst. In many cases, "mass of catalyst" is a good description of the size of the reactor, so we use mass of catalyst as a basis for defining some intensive properties of the system (like a

reaction rate). If you look back at the specific way we defined reaction rates in this example, you'll see that I was careful to include the units of "moles of extent per gram of catalyst per minute." That is an example of an intensive reaction rate normalized to a unit mass of catalyst. We reach a decision about that intensive basis only after considering a sensible description of the size of our system.

1.2.3 Heterogeneous reactions at 2D interfaces

We might imagine that a reaction only occurs at the interface between two phases. A common example might be a reaction between two immiscible liquids. Alternatively, chemical vapor deposition is a widely used technology for synthesizing solid materials. A good example is the chemical vapor deposition of silane to produce silicon. This is a core technology in the manufacture of electronic components.



Here, you have a reaction where the gas phase silane decomposes at the silicon surface to deposit more solid silicon. So, the reaction is only occurring at the solid surface. In this case, the volume of silane is largely irrelevant to the size of the silicon wafer that we are growing. A better way to view the size of this system would be in terms of the surface area of the silicon wafer as the reaction is not occurring anywhere except for the surface of the silicon wafer. Here, we might see reaction rates reported per unit surface area, and we'd find that the total extensive quantity of silane that we're consuming scales with surface area, i.e., interfacial surface area is probably a good basis for the size of this system. Note that we also might use interfacial surface area for heterogeneously catalyzed reactions as the *surface reactions* of heterogeneous catalysis are clearly occurring on 2D surfaces. Finally, areal rates and properties are appropriate for systems that involve immiscible liquids reacting at an interface.

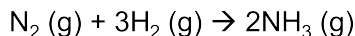
1.2.4 Heterogeneous reactions at exposed surface atoms (active sites)

Catalytic reactions usually don't happen uniformly throughout the mass or volume of a catalyst you put into a reactor. In reality, the catalytic surface has "active sites" where the reaction occurs. Often, these are atoms of expensive metals like Pt, Pd, or Ru. It is far more cost effective to disperse these metals as isolated nanoparticles on an inert, high surface area carrier like SiO_2 or Al_2O_3 than it is to throw chunks of Pt into the reactor (which have a very low percentage of their atoms on the surface—most Pt atoms in a chunk of Pt are inside of the chunk of Pt, which does us no good from a catalysis standpoint. Only the surface atoms are useful.)

So, revising our picture of ammonia synthesis, it probably looks more like this:

Ammonia Synthesis:

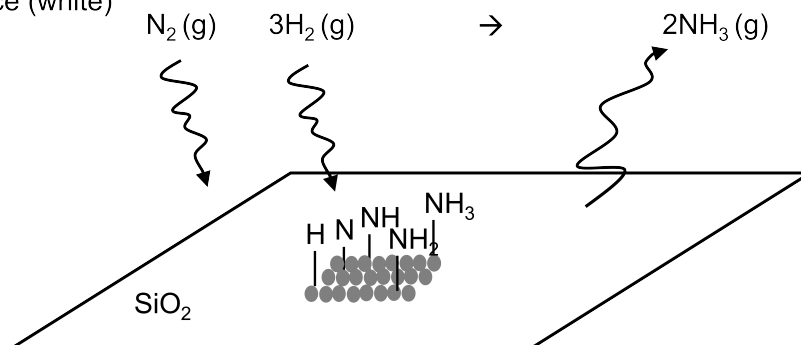
An example of Heterogeneous Reaction



This reaction occurs over Fe-based catalysts

Here is an illustration of Fe nanoparticles (grey)

On an inert SiO_2 surface (white)



In these cases, it isn't the mass of catalyst or even the surface area of catalyst that dictates the extensive performance of the system: it is the number of exposed active sites (Fe atoms in this example, but it could be Pt atoms, Pd atoms, protons, etc). Often, this is the view of a system we take in fundamental catalysis, where we are interested in studying the system on the basis of a single active site, and so we frequently report rates this way (these are usually called turnover frequencies).

1.3 Material Balances on Reacting Systems

As reactions occur, we know that:

1. Total mass is conserved
2. The total number of each element is conserved

However, the nature of that mass changes form as the atoms break apart and recombine in new ways. For example, 1 N_2 and 3 H_2 's become 2 NH_3 's. Our ability to mathematically describe the change in quantities of each species is critical to our ability to design a chemical reactor. It allows us to answer questions like:

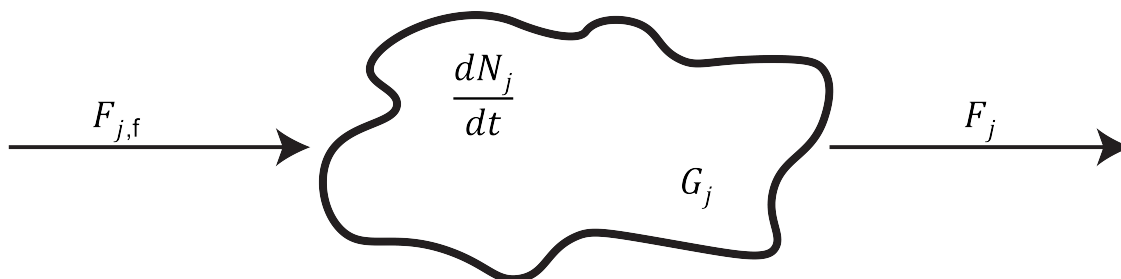
1. What size of reactor do I need to achieve a certain conversion?
2. What conversion will the reactor achieve for a certain mass of catalyst?
3. How much time will it take for the reactor to achieve a certain yield of product?

All of these are **extensive** questions; in other words, the answers to these questions depend on the scale of the system we are considering. This is in contrast to an **intensive** property, which is independent of the scale of the system. To answer these questions, we write **material balances**, which allow us to account for changes in quantities of each species that occur in chemical reactors. Because balanced reactions generally describe changes in molar quantities of reactants and products, we typically write material balances for reactors in molar units. This makes it straightforward for us to use stoichiometry and other properties of reactions that are specified on a molar basis. Material balances (mole balances) on reacting systems are probably the most important concept covered in

a reactor design course. Almost all of our reactor design problems concerned with sizing a process or determining how long a process will take require us to write material balances.

1.3.1 Developing a Material Balance for a Reacting System

To start, we consider a system of arbitrary shape and “size.” Most of us immediately think of Volume when we consider the “size” of a reactor. Volume will always be important in determining the physical space occupied by a reactor as well as its cost. However, as discussed above, volume may not always be the best (mathematical) choice of basis to describe the “size” of our system when reactions occur at interfaces and catalyst surfaces. Before we get into that discussion, let’s look at general aspects of a mole balance that apply for a system of any size or shape.



The squiggly amorphous shape is our reactor – it has an arbitrary “size” and an arbitrary shape. As of now, it is no type of reactor in particular, it just represents a place where a reaction is occurring. If we’re going to write a material balance on that space, we have to consider several things. First, in this course, we usually write material balances for species rather than total material balances. With this in mind, we will develop this balance for an arbitrary species, j . Once we’ve established that we’re going to write a balance on the molar quantity of species j in that squiggly element of arbitrary size and shape, we have to consider a few things that contribute to the balance. One should recognize that each of these things is an extensive quantity because it scales with the size of the system:

1. The accumulation of moles of species j in the reactor as a function of time, dN_j/dt .
2. The “feed” molar flowrate flow of species j into the reactor, $F_{j,f}$.
3. The effluent molar flowrate of species j exiting the reactor, F_j .
4. The rate of generation or consumption of species j by chemical reaction, G_j .

If we can account for those 4 quantities, we fully describe the material balance (mole balance) on species j in our reactor.

I am using the generic notation that N_j is the moles of species j in units of **moles of j** , F_j and $F_{j,f}$ are molar flowrates of species j , with units of **moles of j per time**, and G_j is the extensive rate of “generation” by chemical reaction in units of **moles of j per time**.

We apply the following concept to describe a material balance on a species:

{Accumulation of j } = {Flow of j in} - {Flow of j out} + {Generation of j by chemical reactions}

If we substitute the labelled quantities in the figure above, we arrive at the general material balance equation for a chemical reactor:

$$\frac{dN_j}{dt} = F_{j,f} - F_j + G_j$$

This equation applies (without making assumptions) to any reactor, of any type, of any size, of any shape, and for any number of reactions occurring inside of the reactor. It is a general starting point for writing a balance on *any* reacting system. The units of each term in this equation are *moles per time*. There is no dimensional normalization. A general material balance of this type is therefore an *extensive* description of quantities of species j within and moving through a reactor, and it will change with the size of the system or the type of reactor you are considering.

1.3.2 Analyzing the parts of a material balance on species j

Influent Molar Flowrate Consider the molar flowrate of species j entering the reactor—we usually refer to this as the “feed” molar flowrate of j , and we give it the symbol, $F_{j,f}$. It is an extensive quantity and has units of moles per time. Usually, as engineers—either in industrial practice or in a research lab—we have some control over (or at least some knowledge about) what goes into the reactor. Flowrates can be controlled using a number of devices, and we have numerous ways of either controlling or measuring the composition of an influent stream. Sometimes, we’ll have species molar flowrates (F_j) given to us directly. In other cases, we will be given information about the total molar or volumetric flowrate into the reactor as well as the concentration or composition of each species in that feed stream. For this reason, we will often see component molar flowrates specified in terms of composition (mole fractions) and total molar flowrate, for example:

$$F_{j,f} = y_{j,f} F_{T,f}$$

Where $y_{j,f}$ is the mole fraction of j in the feed, and $F_{T,f}$ is the total molar flowrate of the feed stream. Alternatively, we may be given feed concentrations and a total volumetric flowrate into the reactor. There are practical reasons that we often have access to this type of information – species concentrations are relatively easy to control or measure, and we often control total volumetric flowrates of a fluid phase (using a pump) rather than controlling molar flowrates directly.

$$F_{j,f} = C_{j,f} Q_f$$

Here, $C_{j,f}$ is the concentration of j in the feed stream, and Q_f is the total volumetric flowrate going into the reactor.

Effluent Molar Flowrate Now we consider the molar flowrate of species j leaving the reactor, F_j . More often than not, we will be describing our system in terms of concentration and volumetric flowrates or composition (mole fractions) and total molar flowrates, so we’ll frequently need to relate exit molar flowrates to these quantities. This is done exactly as above, just redefined so that it applies at the “exit” of the reactor:

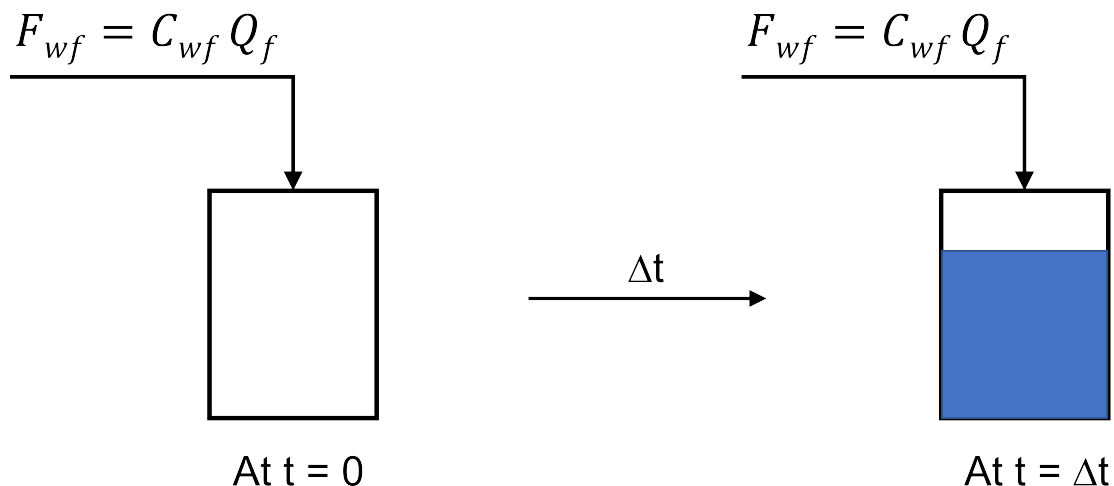
$$F_j = y_j F_T$$

or

$$F_j = C_j Q$$

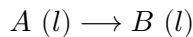
It is worth considering that we will frequently need to measure the composition or concentration of species at the conclusion of the reaction. There is a very simple reason for this: despite knowing the influent composition to a reactor, the reactor is designed to change that composition, and we do not know a priori the extent of that change. As such, the composition exiting the reactor is usually unknown to us, so we must measure it using any number of analytical tools available. Some common techniques for quantifying the composition of a mixture include Gas Chromatography, Liquid Chromatography, NMR, FTIR, and UV-Vis, though the appropriate analytical tool may change depending on the specific system considered. Once we have quantified concentrations and a volumetric flowrate, we can express molar flowrates in terms of these quantities.

Accumulation of species This quantity represents the “accumulation” of species j in the system with respect to time. All this means is that there is a change in the total number of moles of species j in the system as time changes. We can envision “accumulation” in many ways. For example, if we fill a tank with water, we can see clearly that the liquid level and the number of moles of water are changing in that tank as a function of time. This would be an example of accumulation of water in the system, and we can see it pretty clearly from the increase in liquid level!



Clearly, as time goes on, the number of moles of water in the tank are increasing because the flow in is greater than the flow out. This is “accumulation” of water in the tank.

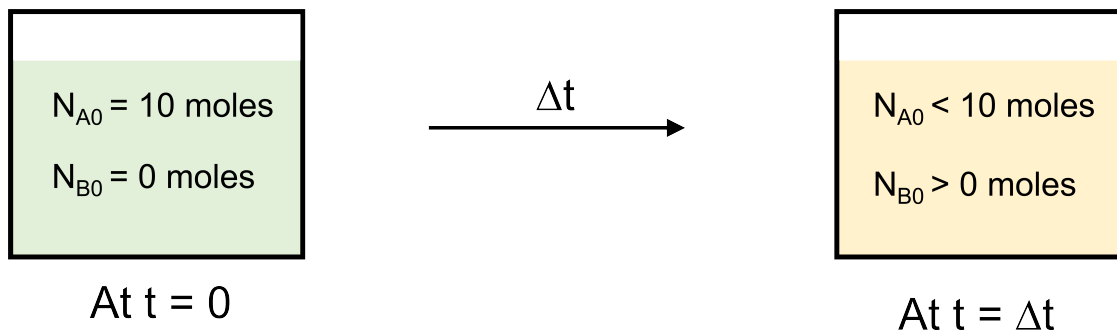
Accumulation without flow A less obvious example of species accumulation is the type of accumulation that occurs in a batch reactor. Batch reactors are closed systems where there is no accumulation of mass; however, the identity of species will change with time. This is perhaps a non-obvious example of accumulation. Consider the reaction:



Let’s say we initially have a tank that will fill with 10 moles of A . It occupies a volume of 1L liter. The density of B is equal to the density of A , so as we convert A into B , the system volume does

not change. Despite this, we have an increase in the number of moles of B in that volume and a decrease in the number of moles of A in that volume as time elapses.

In a batch reactor, total mass is constant. If density is also constant, then the reactor volume must be constant.



We can see clearly that the number of moles of A and B are changing in the tank as a function of time. This means there is “accumulation” of each species despite the fact that there is no flow in or flow out, and the liquid level is constant. All accumulation means in this context is that there is a change in the number of moles in the system as time goes on.

Accumulation can be negative!

The Rate of Generation or Consumption by chemical reaction The final term in the balance is probably the most nuanced. It is the *extensive* rate of generation (or consumption) of species j by chemical reaction. Following the notation in Scott Fogler’s *Elements of Chemical Reaction Engineering*, we will give it the symbol G_j , and it represents how quickly species j is either consumed or produced by chemical reaction(s) that are occurring inside of our reactor. It is definitely related to the rate of reaction, but *it is not the same thing as the rate of reaction*:

$$G_j \neq r_i$$

G_j is a *species specific term*. It is an *extensive* quantity that represents the amount of species j that is produced or consumed by chemical reaction in extensive units of moles-per-time. It will change depending on the size of the system and the particulars of the reactor you are using. In contrast, the rate of reaction, r_i , is a *reaction specific term*. It represents the rate of reaction i that is occurring at the conditions inside of the reactor you are considering. Importantly, in Chemical Reaction Engineering, the rate of a chemical reaction, r_i , is always an *intensive* quantity that is scale and/or system independent. In terms of what controls the intensive rate of reaction, r_i :

$$r_i = f(T, P, \chi_j)$$

To expand that concept: once we decide on what reaction we’re considering and what (if any) catalyst it uses, *the rate of that reaction, r_i , is entirely determined by the temperature, pressure, and composition of the reacting mixture*. The rate of reaction, r_i , does not depend on stoichiometry, nor does it depend on the type, size, or shape of a reactor. It might surprise

you, but the rate of reaction also has no fundamental dependence on important aspects of reactor design, like mixing. The rate of reaction is always entirely determined by temperature, pressure, and composition. In contrast, stoichiometry, shape, size, type of reactor, extent of mixing, etc. will all influence the extensive rate of generation by chemical reaction, G_j :

$$G_j = f(T, P, \chi_j, \text{reactor type, reactor size, mixing, etc.})$$

These two quantities are related, but they are not the same thing. We'll formally develop a generation rate in terms of reaction rates in [Notebook 14](#).